METRIC

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MILITARY SPECIFICATION

BOOT, DUST AND MOISTURE SEAL: HYDRAULIC BRAKE CYLINDER, SYNTHETIC RUBBER

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers brake boots used in covering the open ends of automotive brake cylinders employing hydraulic brake fluid of the nonpetroleum base type.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2530

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SPECIFICATIONS

FEDERAL

VV-B-680 - Brake Fluid, Automotive.

STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.
MIL-STD-45662 - Calibration Systems Requirements.

(Copies of specifications and standards, required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D573 - Rubber-Deterioration in an Air Oven, Test Method for.
ASTM D2000 - Rubber Products in Automotive Applications, Standard Classification System for.
ASTM D2240 - Rubber Property - Durometer Hardness, Test Methods for.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish boots which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information, shall be representative of the units to be furnished to the Government. All subsequent boots delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes (see 6.3).

3.2 Materials. Materials shall be a synthetic, non-oil resistant rubber compound conforming to ASTM D2000 M4AA 510 A13B13 or ASTM D2000 M4AA 610 A13B13 with the following exceptions or additions (see 4.8.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.2.2 Durometer hardness. The durometer hardness of the boot shall be as specified in ASTM D2000 M4AA 510 A13B13 or ASTM D2000 M4AA 610 A13B13 for the applicable rubber compound (see 4.8.3).

3.2.3 Volume change. When tested as specified in 4.8.4, the volume of the test specimen shall neither increase by more than 20 percent (%) nor decrease by more than 5%. After testing, the durometer hardness shall not increase by more than 5 points nor decrease by more than 10 points.

3.2.4 Heat stroking. After being tested as specified in 4.8.5, boots shall be free from flex cracks and shall fit snugly around the cylinder and push rod.

3.2.5 Low temperature flexibility. When tested as specified in 4.8.6, boots shall not crack.

3.2.6 Heat resistance. When tested as specified in 4.8.7, the durometer hardness of the boot shall be not greater than 15 points higher or not more than 5 points lower than the initial durometer hardness. The rubber, after cooling, shall not be tacky and there shall be no cracking when folded.

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3.2.7 Ozone resistance. When tested as specified in 4.8.8, boots shall exhibit no evidence of cracking or rupture under seven power magnification.

3.3 Design and construction. Boots shall be of molded construction and shall conform to the dimensions specified on the applicable drawings (see 4.8.1, 4.8.2, and 6.2).

3.4 Marking. The identification mark of the manufacturer, the capital letters "OZ" and other marking as specified on applicable drawings or by the acquisition activity, shall be molded into each boot (see 4.8.2, 6.2, and 6.4).

3.5 Workmanship. Workmanship shall be such that the defects specified in MIL-STD-407 shall not be present (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 23 ± 10 degrees Celsius (°C).
b. Barometric pressure: 725 + 50 mm Hg.
- 70
c. Relative humidity: 50 ± 30%.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select 10 boots produced under the production contract for first article inspection. First article samples shall be inspected as specified in table I. The applicability of sample number to test and order of testing shall be as specified in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply boots that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

TABLE I. Classification of Inspections.

Title	Requirement	Inspection	First article	Quality conformance		Control
				Examination	Test	
Materials and construction	3.2 and 3.3	4.8.1	X			X
Defects (see 4.5.2.2)	3.3, 3.4, and 3.5	4.8.2	X	X		X
Durometer hardness	3.2.2	4.8.3	X		X	X
Volume change	3.2.3	4.8.4	X			
Heat stroking	3.2.4	4.8.5	X			
Low temperature flexibility	3.2.5	4.8.6	X			1/
Heat resistance	3.2.6	4.8.7	X			
Ozone resistance	3.2.7	4.8.8	X			1/

1/ For control testing, 2 of the 10 boots shall be subjected to the low temperature flexibility test and 2 other boots shall be subjected to the ozone resistance test.

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TABLE II. Applicability of sample number and order of testing.

Test	Sample number									
	1	2	3	4	5	6	7	8	9	10
Durometer hardness	X	X	X	X	X	X	X	X	X	X
Volume change	X	X								
Heat stroking			X	X						
Low temperature flexibility					X	X				
Heat resistance							X	X		
Ozone resistance									X	X

4.4.1 First article inspection failure. Deficiencies found during, or as a result of, the first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government

4.5 Quality conformance inspections.

4.5.1 Sampling.

4.5.1.1 Lot formation. Unless otherwise specified (see 6.2), an inspection lot shall consist of all the boots of one size, design and part number, from an identifiable production period, produced by one manufacturer and submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.1.3 Sampling for tests. Samples for tests shall be selected in accordance with level II of MIL-STD-105.

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4.5.2 Examination.

4.5.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following AQL's, on the basis of percent defective:

Classification	AQL
Major	2.5
Minor	4.0

4.5.2.2 Classification of defects. For examination purposes, defects shall be determined and classified as specified in MIL-STD-407.

4.5.3 Test. Samples selected in accordance with 4.5.1.3 shall be subjected to the quality conformance test specified in table I, and shall conform to an AQL of 2.5.

4.6 Control tests. Control tests samples shall be selected at the rate of 1 sample (consisting of 10 boots) of each 1000 boots produced, except the not less than 1 nor more than 2 samples shall be selected in any 30-day period. Boots shall be inspected as specified in table I, in the order specified herein.

4.7 Failure. Failure of any boot to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 and 3.3, shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3, 3.4, and 3.5 shall be determined in accordance with MIL-STD-407.

4.8.3 Durometer hardness. To determine conformance to 3.2.2, boots shall be tested for durometer hardness in accordance with ASTM D2240.

4.8.4 Volume change. To determine conformance to 3.2.3, two specimens cut from separate boots shall be tested in accordance with method 6211 of FED-STD-601. The test fluid shall conform to VV-B-680. The test fluid temperature shall be 121 ± 3°C. The time of immersion shall be 70 ± 1/4 hours. Each specimen shall be approximately 6.45 square centimeters in area. At the completion of the test, specimens shall be examined to determine the volume change.

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4.8.5 Heat stroking. To determine conformance to 3.2.4, boots shall be tested as follows.

4.8.5.1 Apparatus. Stroking devices shall be provided, consisting of two master cylinder assemblies and two wheel cylinder assemblies or equivalent, on which boots can be mounted with appropriate mechanism to permit cycling as specified herein. An oven, of the type described in ASTM D573 shall be provided.

4.8.5.2 Method. Two boots shall be mounted on the stroking device. The push rods shall then be cycled at the rate of 1000 ± 100 strokes per hour for 120,000 cycles. Stroke length shall be 80% of the maximum stroke for master cylinders and 5 millimeters for wheel cylinders. The ambient air temperature shall be 70 ± 1°C. No oil shall contact the boots during stroking. After the specified cycles have been completed, the boots shall be examined for flex cracks extending through the wall thickness and for fit around cylinder and push rod.

4.8.6 Low temperature flexibility. To determine conformance to 3.2.5, boots shall be tested as follows.

4.8.6.1 Apparatus. Test fixtures shall be provided, consisting of two master cylinder assemblies and two wheel cylinder assemblies or equivalent, on which test boots can be mounted and tested as specified. When boots are designed to slide on the push rods, collars or similar locking devices shall be furnished to prevent relative motion between boot and push rod during test. A cold chamber shall be provided which will contain the test fixture, permit observation and actuation of the fixture mechanism, and maintain a dry air temperature of minus 18.3 ± 2°C.

4.8.6.2 Method. Two boots shall be mounted on the fixture, placed in the cold chamber, and held at a temperature of minus 18.3 ± 2°C for not less than 70 hours. The boots shall be so mounted that relative motion between each boot and push rod will not occur. The push rods shall then be actuated through the maximum stroke, for which the cylinder was designed, for a total of 10 cycles. The boots shall then be removed from the cold chamber and examined for cracks or other damage.

4.8.7 Heat resistance. To determine conformance to 3.2.6, boots shall be tested as follows.

4.8.7.1 Apparatus. A circulating air oven described in ASTM D573 with provisions for suspending boots within the oven shall be provided.

4.8.7.2 Method. Two boots shall be tested. One boot shall be suspended in the oven for 70 ± 1/4 hours at 100 ± 3°C and one boot shall be suspended in the oven for 24 ± 1/4 hours at 121 ± 3°C. After oven aging, the durometer hardness of each boot shall be determined and each boot shall be folded and examined for cracks and for tackiness.

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4.8.8 Ozone resistance. To determine conformance to 3.2.7, two boots shall be assembled on a hydraulic cylinder or equivalent which will provide a 12 to 15% stretch in both the boot-head and push-rod mounting areas. The boots shall then be conditioned in an ozone-free atmosphere at room temperature for not more than 45 minutes. Samples shall then be placed in an ozone chamber with the ozone concentration maintained at 50 ± 5 parts per 100,000,000 parts of air by volume, at a temperature of 38 ± 1°C, for a period of 168 ± 1/2 hours. The boots shall be examined daily for cracking and rupture under seven power magnification. At completion of test, the boots shall be examined again under the same magnification.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Brake boots covered by this specification are intended for use as protective devices against dirt and moisture on wheel and master cylinders in hydraulic brake systems of automobiles, trucks, and other wheeled vehicles employing hydraulic brake fluid of the nonpetroleum base type.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article inspection is not required (see 3.1).
c. Title, number, and date of applicable drawing (see 3.3).
d. Identification marking required (see 3.4).
e. If responsibility for inspection shall be other than as specified (see 4.1).
f. If reponsibility for inspection equipment shall be other than as specified (see 4.1.2).
g. If inspection conditions shall be other than as specified (see 4.3).
h. If first article sample size shall be other than as specified (see 4.4).
i. If inspection lot shall be other than as specified (see 4.5.1.1).
j. Selection of applicable level and packaging requirements (see 5.1).

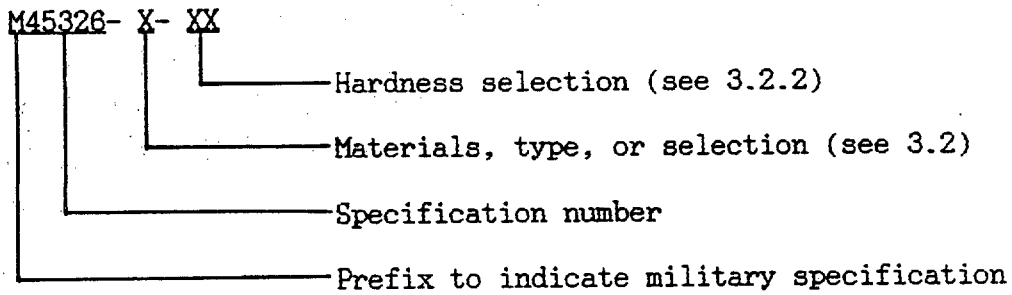
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6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.1.1).

6.3.1.1 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.)(Source: Federal Acquisition Regulations, section 23.402).

6.4 Part or Identifying Number (PIN). The PIN to be used for boots acquired to this specification are created as follows:



6.5 Subject term (key word) listing.

Hardness
Heat stroking
Low temperature flexibility
Nonpetroleum base fluid
Non-oil resistant rubber
Ozone resistance
Volume change

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AT

Review activities:
Army - MI
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Preparing activity:
Army - AT

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-B-45326C(AT); Boot, Dust and Moisture Seal: Hydraulic Brake Cylinder, Synthetic Rubber

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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